

From the Forehead to the Ear: How We Redesigned the NeuroFocus Electrode Mount

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NeuroFocus measures focus and fatigue signals from competitive gamers through hardware that lives inside the headset they already wear. That one constraint — “inside the headset they already wear” — has driven every hardware decision we’ve made. This is the story of how our electrode mount went from the forehead to the ear, and what we learned building both.

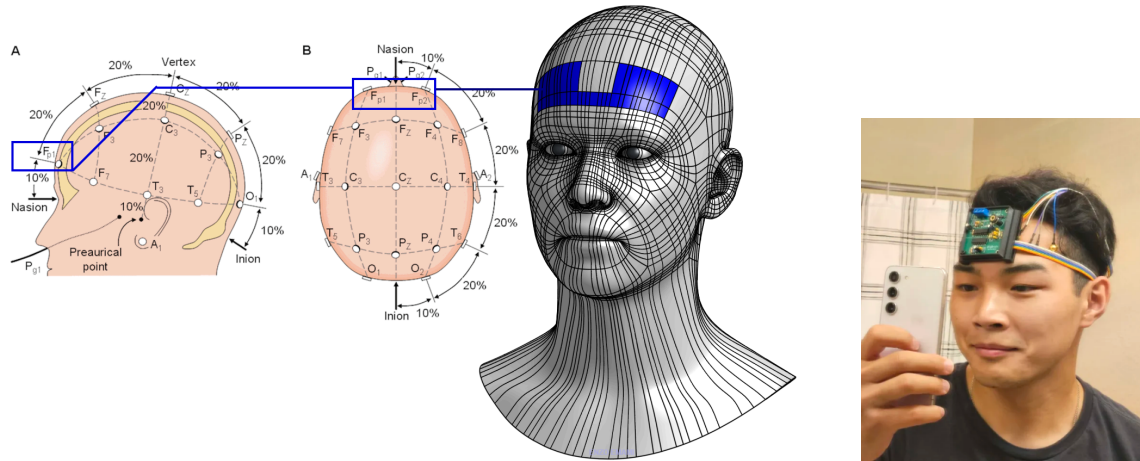
The problem: EEG needs skin contact; gamers need comfort

An EEG electrode is only as good as its contact with the scalp. Too little pressure and you get noise and dropouts; too much and the player is adjusting their headset mid-game — which is both a comfort failure and a data failure, since every readjustment is an artifact in the signal. Our job on the hardware side was to hold an electrode against skin at a consistent, tunable pressure, across wildly different head shapes, inside headsets we don't manufacture.

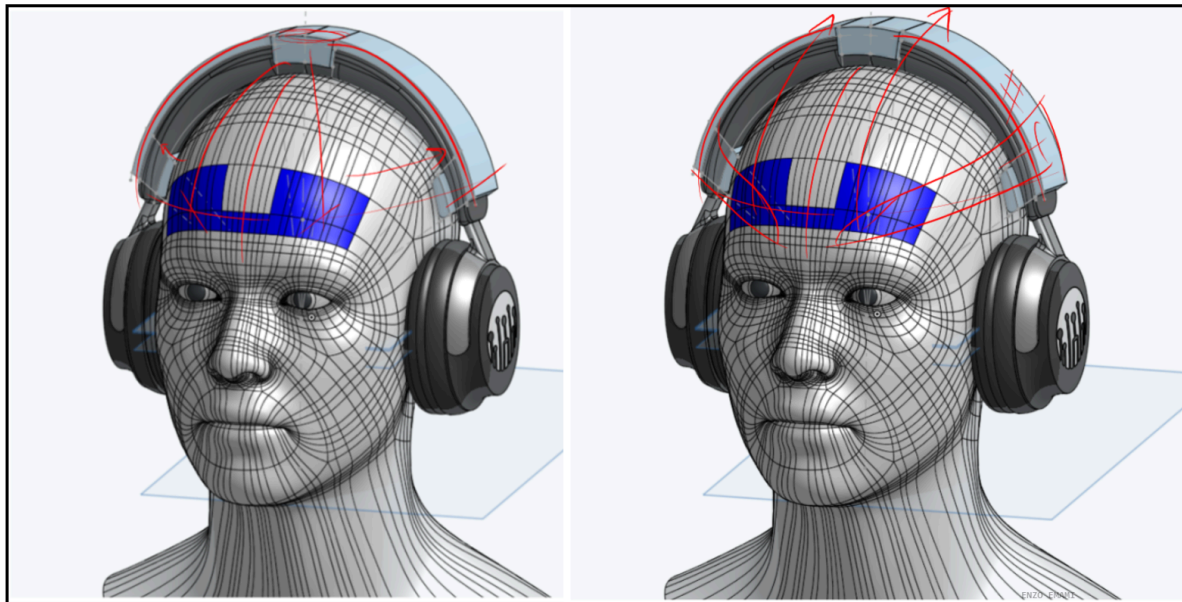


Chapter 1 — The frontopolar approach (FP1/FP2)

Our first architecture targeted the frontopolar positions — FP1 and FP2, on the forehead just above the eyebrows. These are classic sites for attention-related EEG work, and the forehead is hairless, which makes dry-electrode contact much easier.



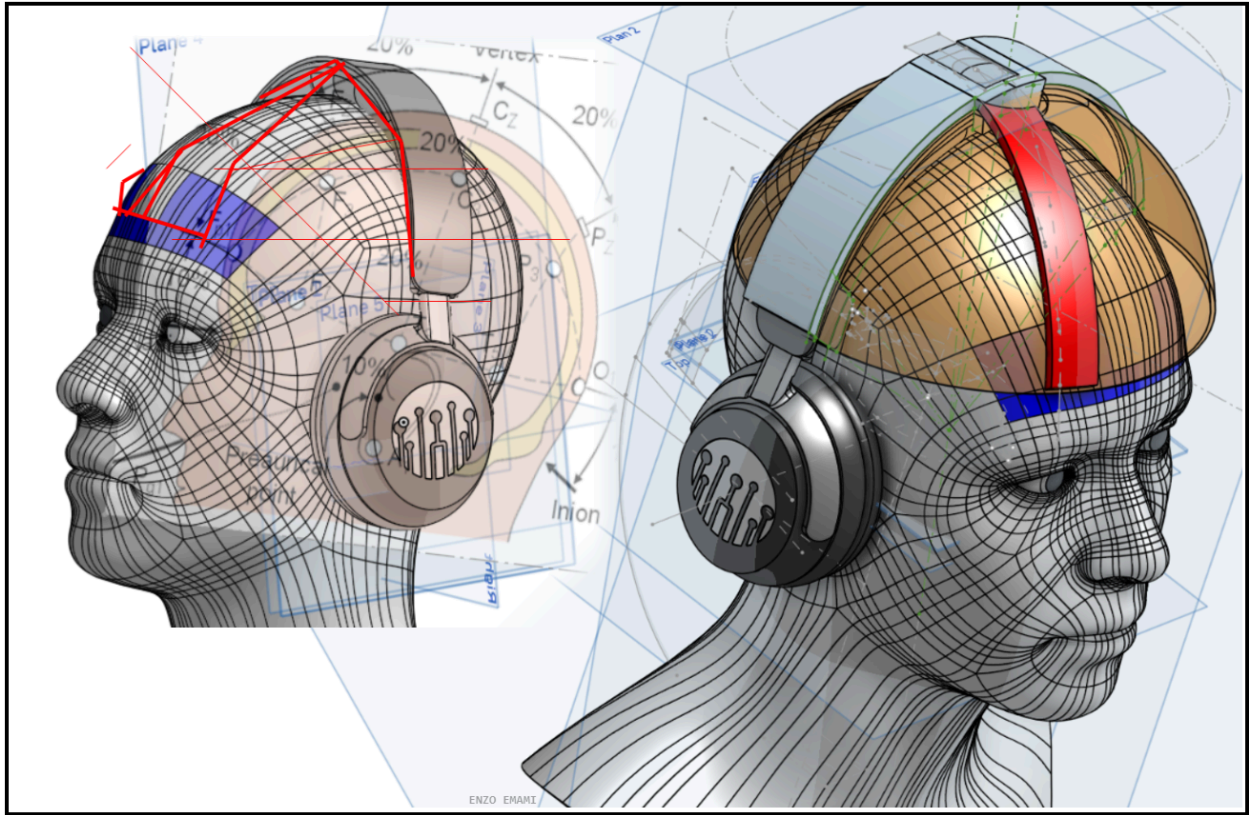
The catch: gaming headsets don't touch your forehead. So the mount had to bridge from the headband down to the forehead, and it had to do so on any headset — different band widths, curvatures, clamping forces, and padding thicknesses.



The vertical approach

- Our first mount concept used a single pivot point at the top of the headband. Two arms reached down to FP1 and FP2, and when not in use, they swiveled up and tucked into the arc of the headset — invisible until you needed them.

- The arms were designed as a single TPU print with tuned zones: dense infill at the top for a rigid pivot base, transitioning to a flexible section along the arm so the tip could flex toward the head. Rather than guessing at contact pressure, I created a compression-angle variable that drove the arm geometry — regenerate at a new angle, print, test comfort, repeat.

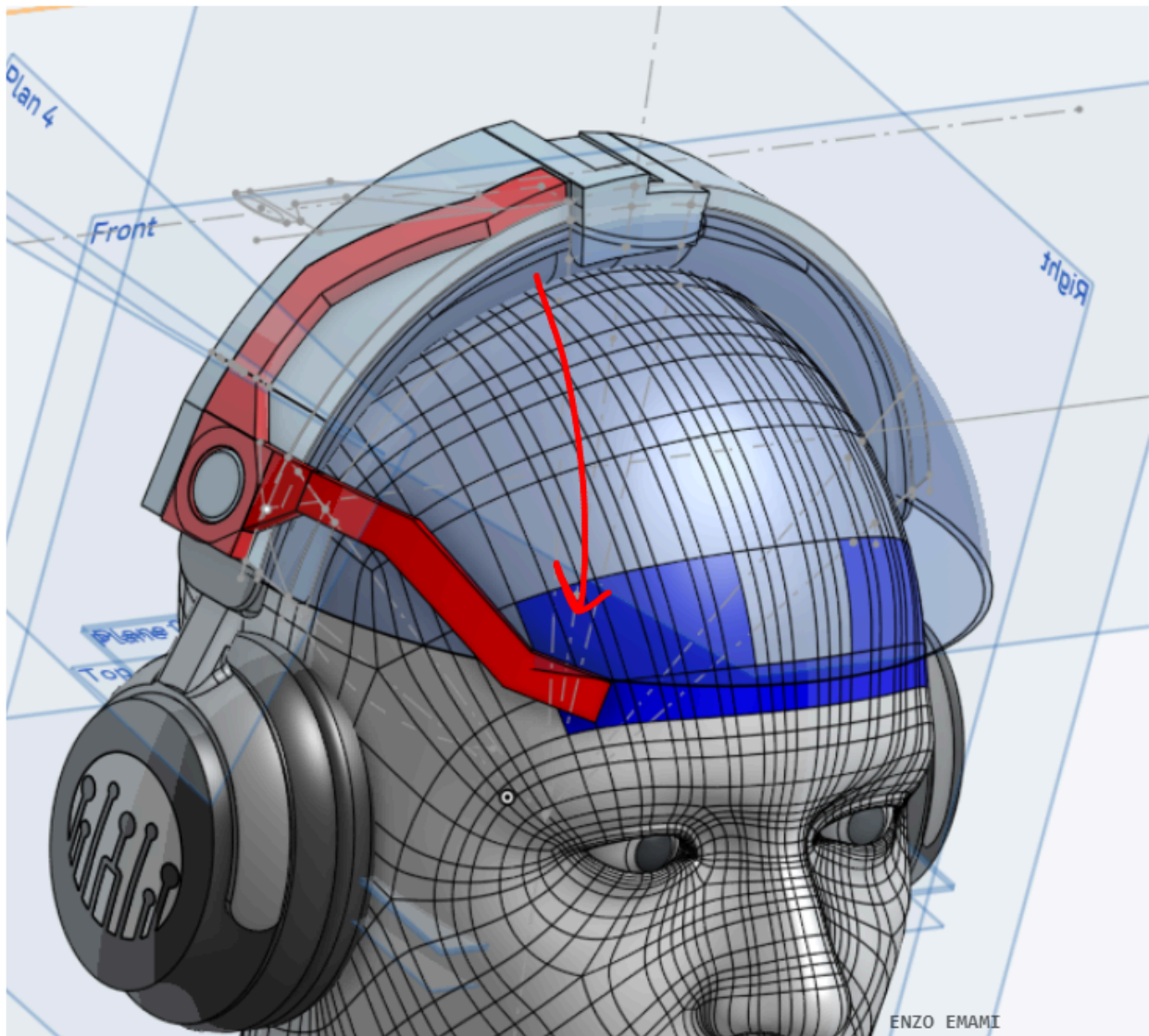


Caption: The single-pivot vertical mount — arms deployed to FP1/FP2 (left) and swiveled into the headband arc for storage (right).

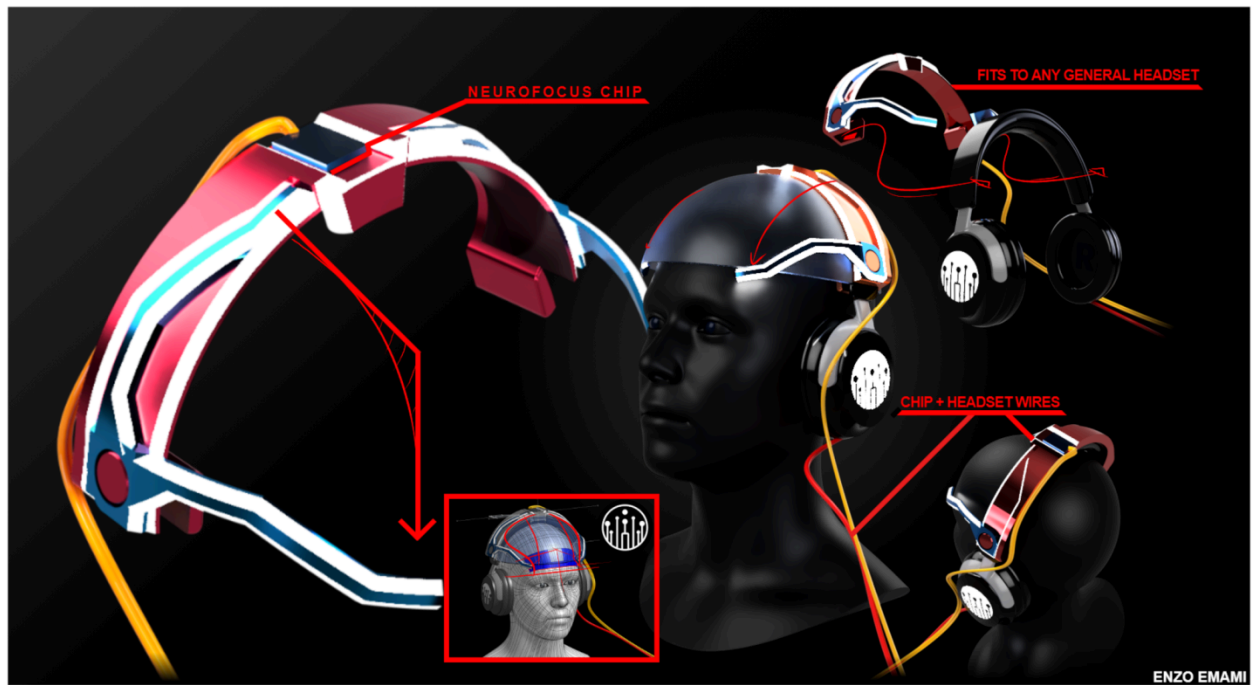
The horizontal approach

The vertical concept worked, but stacking two pivots in the same crowded space at the top of the band made it bulky. So we rotated the whole idea: pivot points moved to the sides, with one

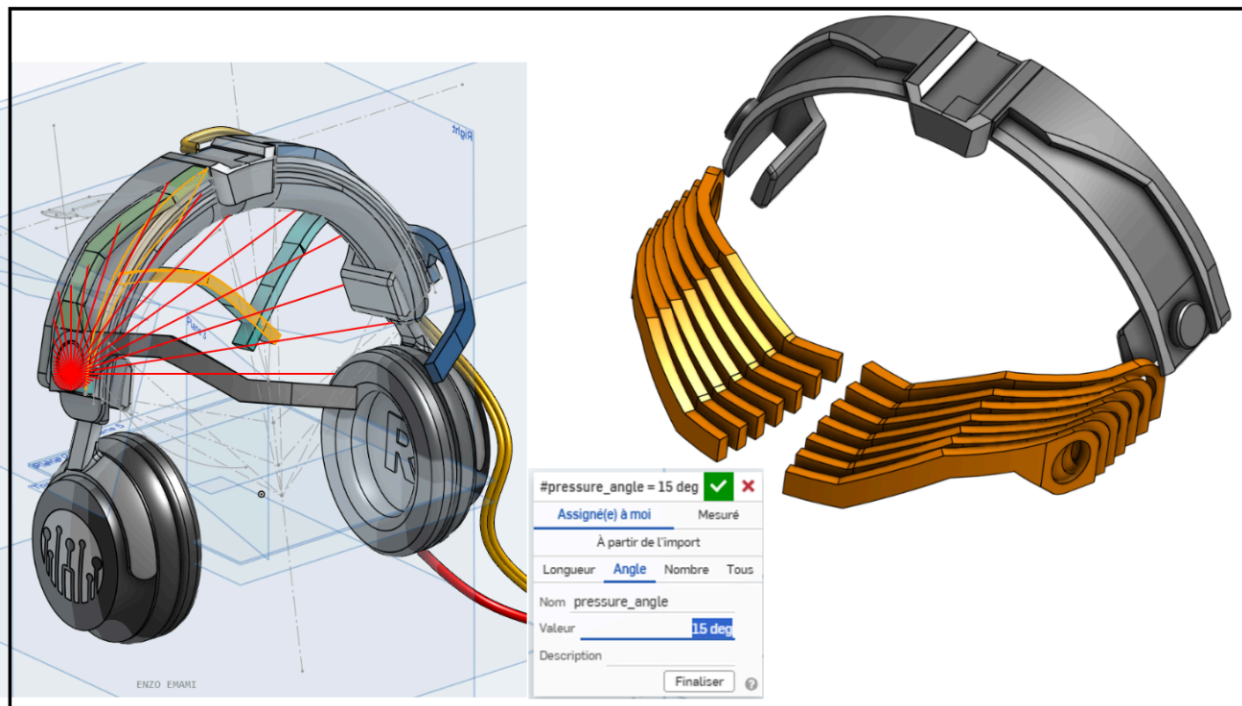
arm reaching in from each side toward the forehead.



This unlocked two things at once. The top of the headband was now free, so the chip could sit there — and without two pivot mechanisms fighting for the same volume, the entire assembly got thinner and lighter. The parametric setup grew with it: new variables for reach and compression angle on each arm.



To find the right contact pressure, we iterated seven pressure angles from 1° to 15° and compared comfort across them.

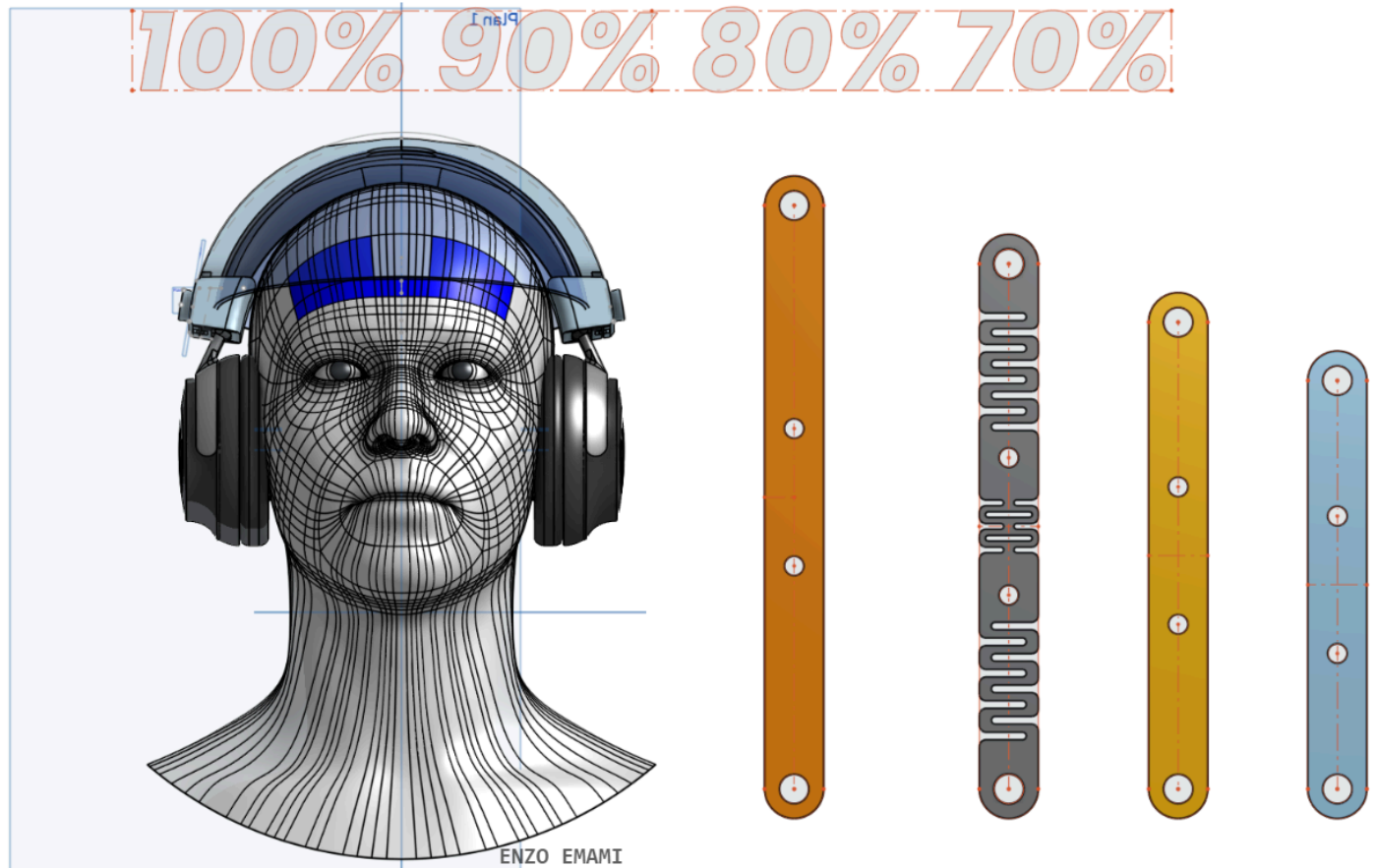


We then manufactured TPU prints of the horizontal V1 and put them on real heads.

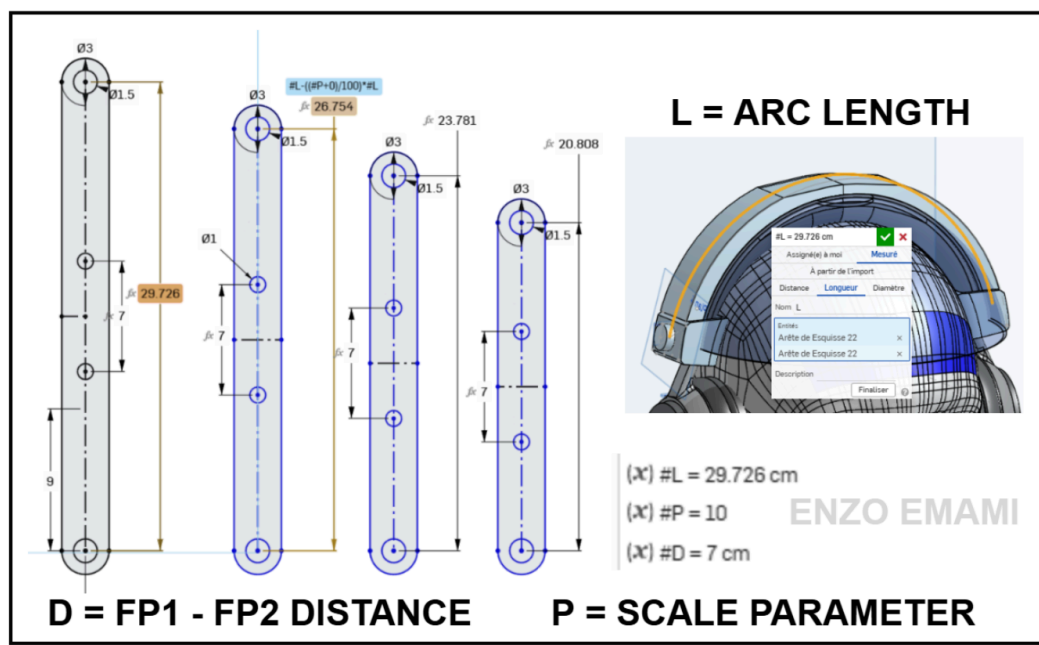


Horizontal V2 — the band

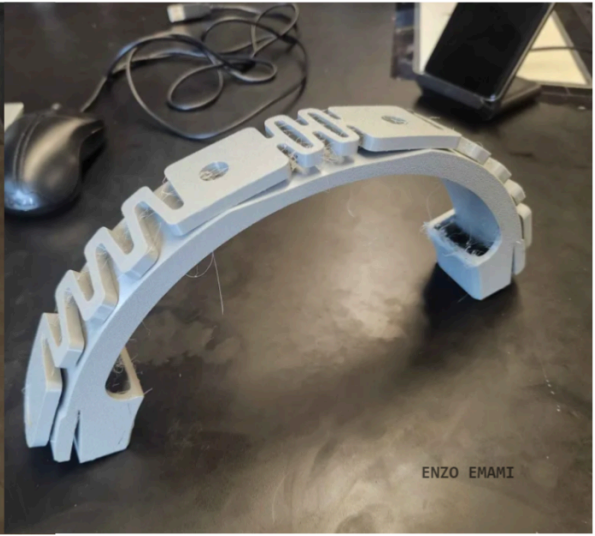
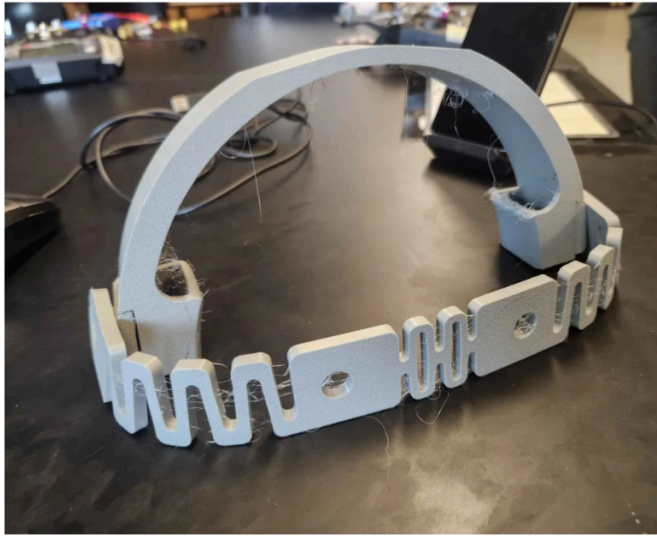
Testing exposed two problems: the arms didn't have quite enough reach, and they wouldn't hold reliably on the same spot session after session. The fix reframed the design — instead of two independent arms, why not connect them into a single band spanning the forehead?



V2 kept the two side pivots (so the band still folds up into the headset's arc when not in use) but joined the arms with an extensible zigzag pattern that stretches across head sizes while holding position. I parametrized the zigzag into different compression percentages, so tuning fit became — once again — a table edit rather than a remodel.

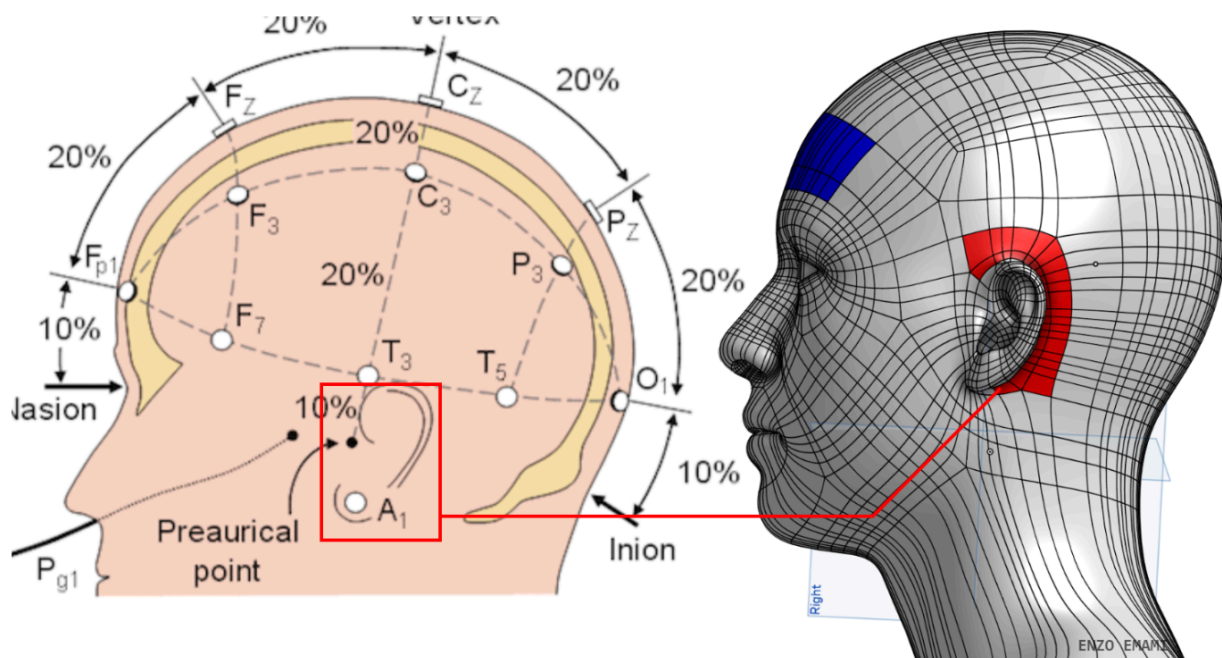






Chapter 2 — Why we moved to the ear

The frontopolar mount worked, but it fought the product. Every headset needed a fit check; the forehead arm was visible (gamers on stream care about that), and the bridge structure added a failure point. Meanwhile, the ear pad is the one place where every gaming headset already presses firmly and consistently against the head.

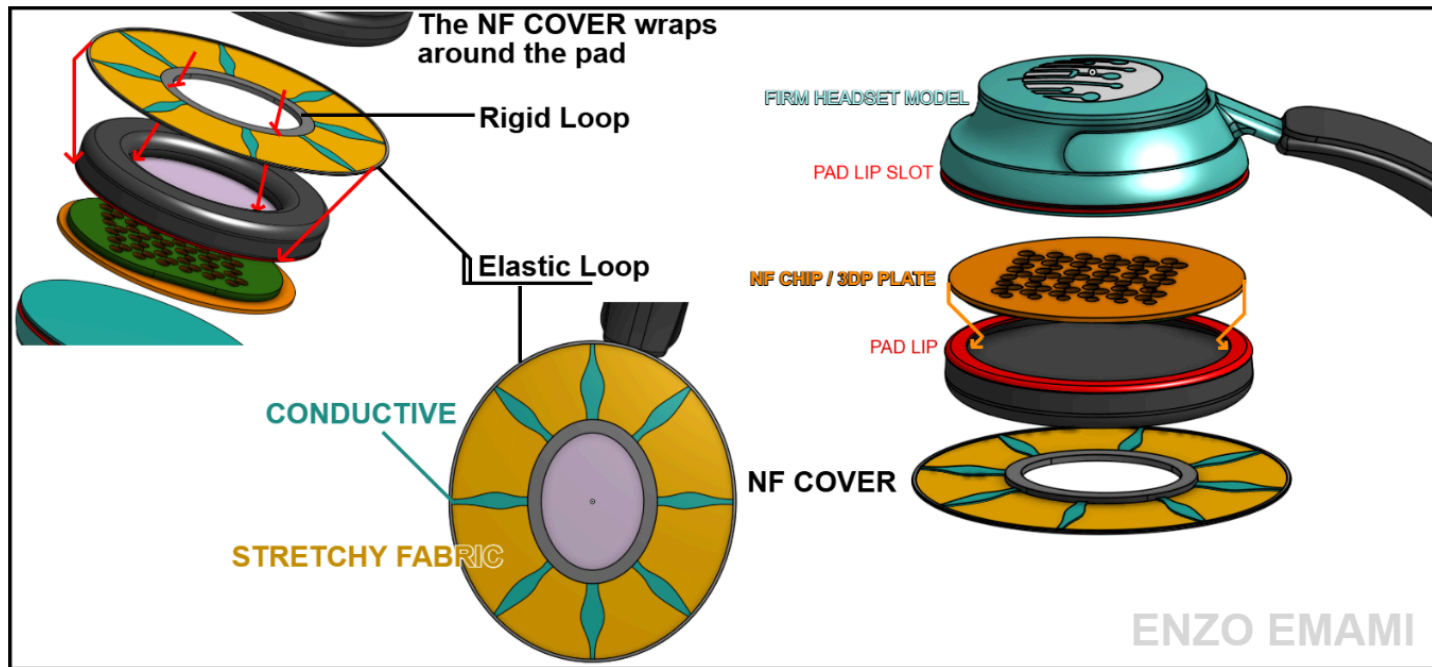


So we pivoted the electrode to the area around the ear, integrating the sensing surface into the ear pad itself using a concentric-ring electrode topology. The headset's own clamping force — a force headset manufacturers already engineer carefully for comfort — became our contact pressure mechanism. Instead of building a structure to create pressure, we borrow pressure that's already there.

Chapter 3 — The ear-pad integration (“the sock”)

The new design wraps the ear pad in what we internally call the sock: a fabricated sleeve that carries the concentric ring electrode and its lead, slipping over a stock ear cushion. It preserves the pad's comfort and acoustics while placing the flexible AND conductive surface exactly where the pad meets skin.

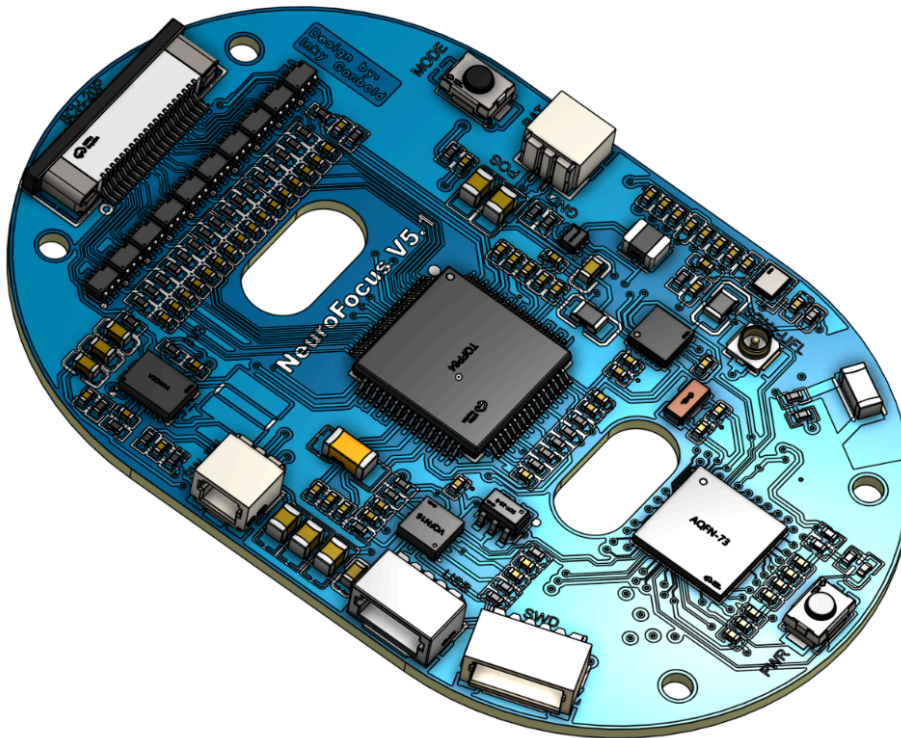
- Works with stock ear pads — no custom cushion tooling required at this stage.
- The concentric ring geometry keeps contact through small headset shifts, which matters over multi-hour sessions.
- The parametric habits carried over: sleeve dimensions, stretch allowances, and ring placement are all variable-driven, so adapting to a different pad size is a table edit.



Silver-plated stretch fabrics run under ~ 1 ohm per square of surface resistivity, some as low as $< 0.2\text{--}0.5 \Omega/\text{sq}$, and the resistance barely moves when stretched (one maker measured $\sim 10\%$ change from relaxed to fully stretched). So a band running to the connector adds negligible series resistance.

What carried over

Almost none of the frontopolar geometry survived the pivot — and almost all of the methodology did. The parametric discipline (pressure as a variable, adaptation as a table edit, print settings tied to geometry) is exactly what let us stand up the auricular design quickly. The forehead mount wasn't wasted work; it was the version that taught us what the real constraints were.



We will continue updating here as we progress.
Thanks for reading ^^

— Enzo, hardware & product design, NeuroFocus.
Last Update: July 11, 2026